

A Polynomial Parametrization of the Surface

$$x^3 + yz + 1 = 0$$

Abstract

We construct an explicit two-parameter polynomial family of integral solutions to

$$x^3 + yz + 1 = 0.$$

More precisely, the polynomial map

$$(a, b) \mapsto (ab - 1, a, -b(a^2b^2 - 3ab + 3))$$

takes $\mathbb{Z}^2$ into the set of integral solutions. Over $\mathbb{Q}$, the same map is a bijection onto the full set of rational solutions. We prove these assertions, characterize exactly which integral points arise from integral parameter values, record two useful one-parameter families, and explain in detail the factorization and parameter-splitting idea that leads naturally to the construction.

1 The polynomial construction

For a commutative ring R , write

$$S(R) = \{(x, y, z) \in R^3 : x^3 + yz + 1 = 0\}.$$

The elementary identity underlying the construction is

$$X^3 + 1 = (X + 1)(X^2 - X + 1). \tag{1}$$

Theorem 1 (Two-parameter polynomial family). *For arbitrary $a, b \in \mathbb{Z}$, define*

$$\boxed{\begin{aligned} x &= ab - 1, \\ y &= a, \\ z &= -b(a^2b^2 - 3ab + 3). \end{aligned}} \tag{2}$$

Then $(x, y, z) \in S(\mathbb{Z})$. Equivalently, the following identity holds in the polynomial ring $\mathbb{Z}[a, b]$:

$$(ab - 1)^3 + a[-b(a^2b^2 - 3ab + 3)] + 1 = 0.$$

Proof. Set $X = ab - 1$. Then

$$X + 1 = ab$$

and

$$\begin{aligned} X^2 - X + 1 &= (ab - 1)^2 - (ab - 1) + 1 \\ &= a^2b^2 - 3ab + 3. \end{aligned}$$

Applying (1) gives

$$X^3 + 1 = ab(a^2b^2 - 3ab + 3).$$

On the other hand, the definitions of y and z give

$$yz = a[-b(a^2b^2 - 3ab + 3)] = -ab(a^2b^2 - 3ab + 3).$$

Hence $X^3 + yz + 1 = 0$. Since all three coordinates in (2) are polynomials with integer coefficients, they are integers whenever a and b are integers. $\square$

Corollary 2 (Infinitely many integral solutions). *For every $t \in \mathbb{Z}$,*

$$\boxed{(x, y, z) = (t, t + 1, -(t^2 - t + 1))} \quad (3)$$

is an integral solution. These solutions are pairwise distinct as t varies; in particular, $S(\mathbb{Z})$ is infinite.

Proof. In Theorem 1, take $a = t + 1$ and $b = 1$. Then $x = (t + 1) - 1 = t$, while

$$a^2 - 3a + 3 = (t + 1)^2 - 3(t + 1) + 3 = t^2 - t + 1,$$

which gives (3). Distinct values of t give distinct values of the first coordinate $x = t$. $\square$

2 Completeness over the rational numbers

The polynomial family above is not merely a convenient source of solutions. With rational parameters it describes every rational point on the surface.

Theorem 3 (All rational solutions). *Define*

$$\Phi : \mathbb{Q}^2 \longrightarrow S(\mathbb{Q}), \quad \Phi(a, b) = (ab - 1, a, -b(a^2b^2 - 3ab + 3)).$$

Then Φ is a bijection. Its inverse is given explicitly by

$$\Phi^{-1}(x, y, z) = \begin{cases} \left(y, \frac{x+1}{y}\right), & y \neq 0, \\ \left(0, -\frac{z}{3}\right), & y = 0. \end{cases} \quad (4)$$

Proof. Theorem 1 proves the defining relation as a polynomial identity, so it remains valid for $a, b \in \mathbb{Q}$. Thus Φ is well-defined.

Let $(x, y, z) \in S(\mathbb{Q})$. First suppose that $y \neq 0$, and put

$$a = y, \quad b = \frac{x+1}{y}.$$

Then $ab = x + 1$, and hence $ab - 1 = x$. Moreover,

$$\begin{aligned} a^2b^2 - 3ab + 3 &= (x+1)^2 - 3(x+1) + 3 \\ &= x^2 - x + 1. \end{aligned}$$

Consequently,

$$\begin{aligned} -b(a^2b^2 - 3ab + 3) &= -\frac{x+1}{y}(x^2 - x + 1) \\ &= -\frac{x^3 + 1}{y} \\ &= z, \end{aligned}$$

where the last equality is exactly the equation $x^3 + yz + 1 = 0$. Hence $\Phi(a, b) = (x, y, z)$.

Now suppose that $y = 0$. The defining equation becomes $x^3 + 1 = 0$. Over $\mathbb{Q}$ this forces $x = -1$. Taking

$$a = 0, \quad b = -\frac{z}{3},$$

we obtain

$$\Phi(a, b) = (-1, 0, -3b) = (-1, 0, z).$$

Thus Φ is surjective and (4) is a right inverse.

To prove injectivity, suppose $\Phi(a, b) = \Phi(a', b')$. Equality of the second coordinates gives $a = a'$. If this common value is nonzero, equality of the first coordinates gives $ab = a'b'$, hence $b = b'$. If $a = a' = 0$, equality of the third coordinates gives $-3b = -3b'$, hence again $b = b'$. Therefore $(a, b) = (a', b')$, and Φ is bijective. $\square$

Proposition 4 (The image of integral parameter values). *Let $(x, y, z) \in S(\mathbb{Z})$. Then $(x, y, z) \in \Phi(\mathbb{Z}^2)$ if and only if one of the following mutually exclusive conditions holds:*

1. $y \neq 0$ and $y \mid x + 1$;
2. $y = 0$ and $3 \mid z$.

In the first case the unique integral parameters are

$$(a, b) = \left(y, \frac{x + 1}{y} \right),$$

and in the second case they are

$$(a, b) = \left(0, -\frac{z}{3} \right).$$

Proof. By Theorem 3, every point of $S(\mathbb{Z})$ has the unique rational parameters displayed in (4). If $y \neq 0$, those parameters are integral exactly when $(x + 1)/y \in \mathbb{Z}$, which is equivalent to $y \mid x + 1$. If $y = 0$, the equation forces $x = -1$, and the unique parameters are integral exactly when $-z/3 \in \mathbb{Z}$, which is equivalent to $3 \mid z$. $\square$

Remark 5. The distinction in Proposition 4 is essential. For example, $(2, 9, -1)$ is an integral solution because $2^3 + 9(-1) + 1 = 0$, but its unique parameters are $(9, 1/3)$, not an integer pair. Thus (2) gives a large polynomial family of integral points but does not exhaust $S(\mathbb{Z})$ when the parameters are required to be integral.

3 How the construction was found

The construction is guided by two requirements: the equation should hold identically, and every displayed coordinate should remain a polynomial in freely chosen parameters. The following sequence of observations leads directly to (2).

1. Isolate the product

The equation may be rewritten as

$$yz = -(x^3 + 1). \tag{5}$$

Thus one does not need to control y and z separately at first; one only needs to split the polynomial $-(x^3 + 1)$ into two factors.

2. Use the sum-of-cubes factorization

The cubic has the canonical factorization

$$x^3 + 1 = (x + 1)(x^2 - x + 1).$$

The most immediate choice in (5) is therefore

$$y = x + 1, \quad z = -(x^2 - x + 1).$$

This already gives the one-parameter family

$$(x, y, z) = (t, t + 1, -(t^2 - t + 1)).$$

It is correct, but it uses only one free parameter.

3. Split the linear factor to create a second parameter

To introduce a second independent parameter without creating fractions, the linear factor $x + 1$ is the natural place to intervene. Write

$$x + 1 = ab. \tag{6}$$

This equation can be solved polynomially for x :

$$x = ab - 1.$$

We then distribute the two factors a and b between y and z :

$$y = a, \quad z = -b(x^2 - x + 1).$$

With these choices,

$$yz = -ab(x^2 - x + 1) = -(x + 1)(x^2 - x + 1) = -(x^3 + 1),$$

so the original equation is forced.

Finally, substituting $x = ab - 1$ into the remaining quadratic factor gives

$$x^2 - x + 1 = (ab - 1)^2 - (ab - 1) + 1 = a^2b^2 - 3ab + 3.$$

This is precisely the polynomial appearing in (2).

4. Why this split is particularly efficient

One could try to split the quadratic factor instead, for example by imposing $x^2 - x + 1 = ab$. However, solving that relation for x would normally require extracting a square root, since

$$(2x - 1)^2 = 4ab - 3.$$

That approach therefore introduces an additional square condition and does not immediately produce polynomial coordinates. By contrast, $x + 1 = ab$ is linear in x , so eliminating x preserves polynomiality and costs no auxiliary condition. The useful design principle is: *split a factor that is linear in the original variable, then absorb the complementary factor into the remaining product variable.*

5. Why rational completeness is visible from the same idea

Suppose a rational solution has $y \neq 0$. Since the construction sets $a = y$ and requires $ab = x + 1$, the only possible value of the second parameter is

$$b = \frac{x + 1}{y}.$$

The original equation then forces the z -coordinate to equal the one produced by the parametrization. Thus the factor-splitting argument not only creates solutions; it also recovers every rational solution away from $y = 0$. The remaining locus $y = 0$ is the line $(-1, 0, z)$, and the specialization $a = 0$ gives $\Phi(0, b) = (-1, 0, -3b)$, which covers that line over $\mathbb{Q}$ as well.

4 A second one-parameter family

There is also a symmetric-looking family obtained from a different factorization pattern.

Proposition 6. *For every $t \in \mathbb{Z}$,*

$$(x, y, z) = (-t^2, t^3 - 1, t^3 + 1) \tag{7}$$

is an integral solution of $x^3 + yz + 1 = 0$.

Proof. Using the difference-of-squares factorization,

$$yz = (t^3 - 1)(t^3 + 1) = t^6 - 1.$$

Since $x = -t^2$, one has $x^3 = -t^6$. Therefore

$$x^3 + yz + 1 = -t^6 + (t^6 - 1) + 1 = 0.$$

□

Remark 7 (Intuition for the second family). To find (7), choose $x = -t^2$ so that $-x^3 = t^6$ is a perfect square and

$$-(x^3 + 1) = t^6 - 1.$$

The right-hand side factors as $(t^3 - 1)(t^3 + 1)$, which can be assigned directly to y and z . This illustrates the same general strategy: make $-(x^3 + 1)$ factor in a parameter-controlled way and assign its factors to the product yz .

5 Conclusion

The principal polynomial parametrization is

$$(x, y, z) = (ab - 1, a, -b(a^2b^2 - 3ab + 3)), \quad a, b \in \mathbb{Z}.$$

Its validity is an identity in $\mathbb{Z}[a, b]$, so no congruence, divisibility, or exceptional-case condition is required for integral parameter values. Over $\mathbb{Q}$, the same map is a bijection onto the entire rational solution set, with inverse given by (4). The construction follows naturally from factoring $x^3 + 1$ and splitting its linear factor $x + 1$ into two free parameters.